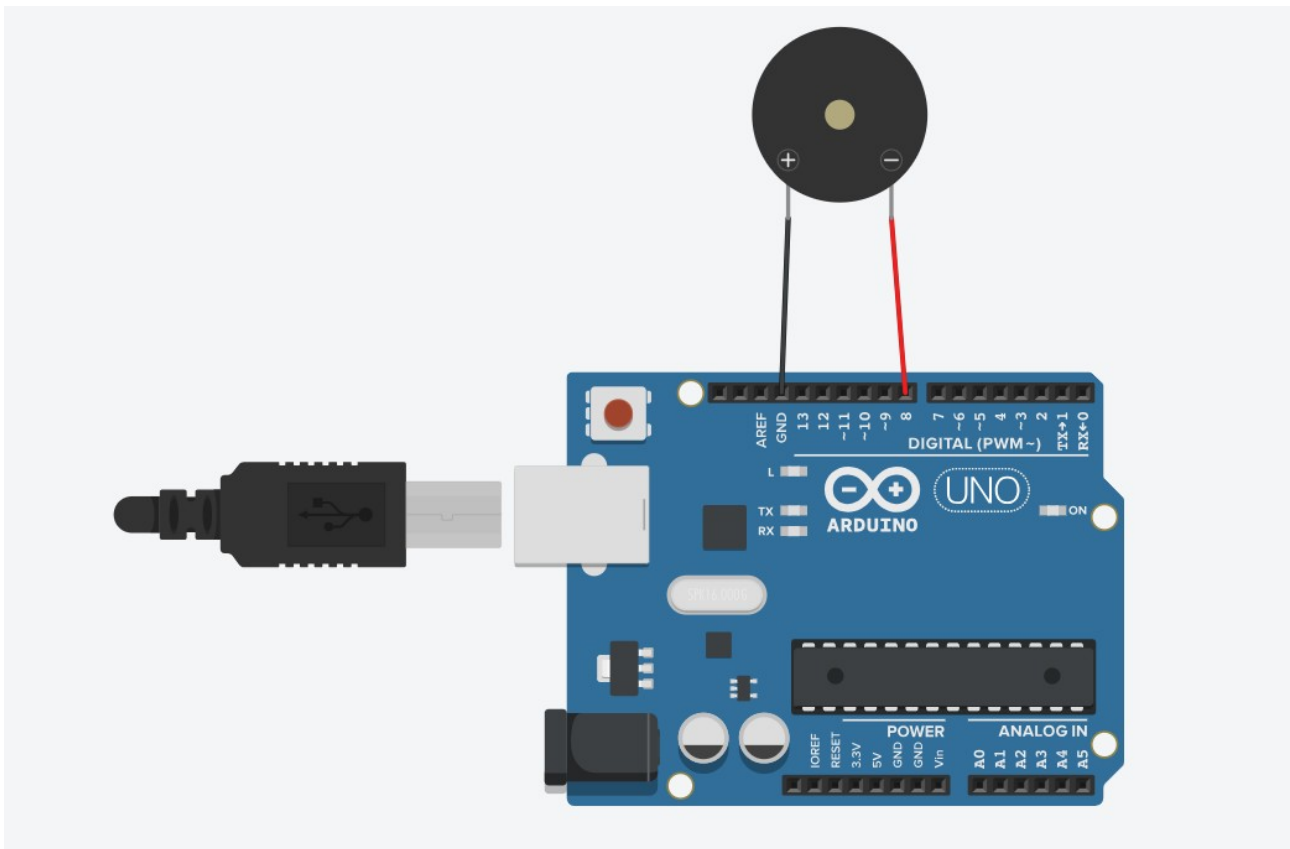


Álvaro García Zafra
P5 EC

Zumbador



```
/******
```

```
Public Constants
```

```
*****/
```

```
#define NOTE_C4 262
#define NOTE_D4 294
#define NOTE_E4 330
#define NOTE_F4 349
#define NOTE_G4 392
#define NOTE_A4 440
#define NOTE_B4 494
#define NOTE_C5 523
#define NOTE_D5 587
#define NOTE_E5 659
#define NOTE_F5 698
#define NOTE_G5 784
#define NOTE_A5 880
#define NOTE_B5 988
```

```
// notes in the melody:
```

```
int melody[] = {
  NOTE_E4, NOTE_G4, NOTE_A4, NOTE_A4, 0,
  NOTE_A4, NOTE_B4, NOTE_C5, NOTE_C5, 0,
```

NOTE_C5, NOTE_D5, NOTE_B4, NOTE_B4, 0,
NOTE_A4, NOTE_G4, NOTE_A4, 0,

NOTE_E4, NOTE_G4, NOTE_A4, NOTE_A4, 0,
NOTE_A4, NOTE_B4, NOTE_C5, NOTE_C5, 0,
NOTE_C5, NOTE_D5, NOTE_B4, NOTE_B4, 0,
NOTE_A4, NOTE_G4, NOTE_A4, 0,

NOTE_E4, NOTE_G4, NOTE_A4, NOTE_A4, 0,
NOTE_A4, NOTE_C5, NOTE_D5, NOTE_D5, 0,
NOTE_D5, NOTE_E5, NOTE_F5, NOTE_F5, 0,
NOTE_E5, NOTE_D5, NOTE_E5, NOTE_A4, 0,

NOTE_A4, NOTE_B4, NOTE_C5, NOTE_C5, 0,
NOTE_D5, NOTE_E5, NOTE_A4, 0,
NOTE_A4, NOTE_C5, NOTE_B4, NOTE_B4, 0,
NOTE_C5, NOTE_A4, NOTE_B4, 0,

NOTE_A4, NOTE_A4,

//Repeat of first part

NOTE_A4, NOTE_B4, NOTE_C5, NOTE_C5, 0,
NOTE_C5, NOTE_D5, NOTE_B4, NOTE_B4, 0,
NOTE_A4, NOTE_G4, NOTE_A4, 0,

NOTE_E4, NOTE_G4, NOTE_A4, NOTE_A4, 0,
NOTE_A4, NOTE_B4, NOTE_C5, NOTE_C5, 0,
NOTE_C5, NOTE_D5, NOTE_B4, NOTE_B4, 0,
NOTE_A4, NOTE_G4, NOTE_A4, 0,

NOTE_E4, NOTE_G4, NOTE_A4, NOTE_A4, 0,
NOTE_A4, NOTE_C5, NOTE_D5, NOTE_D5, 0,
NOTE_D5, NOTE_E5, NOTE_F5, NOTE_F5, 0,
NOTE_E5, NOTE_D5, NOTE_E5, NOTE_A4, 0,

NOTE_A4, NOTE_B4, NOTE_C5, NOTE_C5, 0,
NOTE_D5, NOTE_E5, NOTE_A4, 0,
NOTE_A4, NOTE_C5, NOTE_B4, NOTE_B4, 0,
NOTE_C5, NOTE_A4, NOTE_B4, 0,

//End of Repeat

NOTE_E5, 0, 0, NOTE_F5, 0, 0,
NOTE_E5, NOTE_E5, 0, NOTE_G5, 0, NOTE_E5, NOTE_D5, 0, 0,
NOTE_D5, 0, 0, NOTE_C5, 0, 0,
NOTE_B4, NOTE_C5, 0, NOTE_B4, 0, NOTE_A4,

NOTE_E5, 0, 0, NOTE_F5, 0, 0,
NOTE_E5, NOTE_E5, 0, NOTE_G5, 0, NOTE_E5, NOTE_D5, 0, 0,
NOTE_D5, 0, 0, NOTE_C5, 0, 0,
NOTE_B4, NOTE_C5, 0, NOTE_B4, 0, NOTE_A4};

```

int duration = 500; // 500 milliseconds

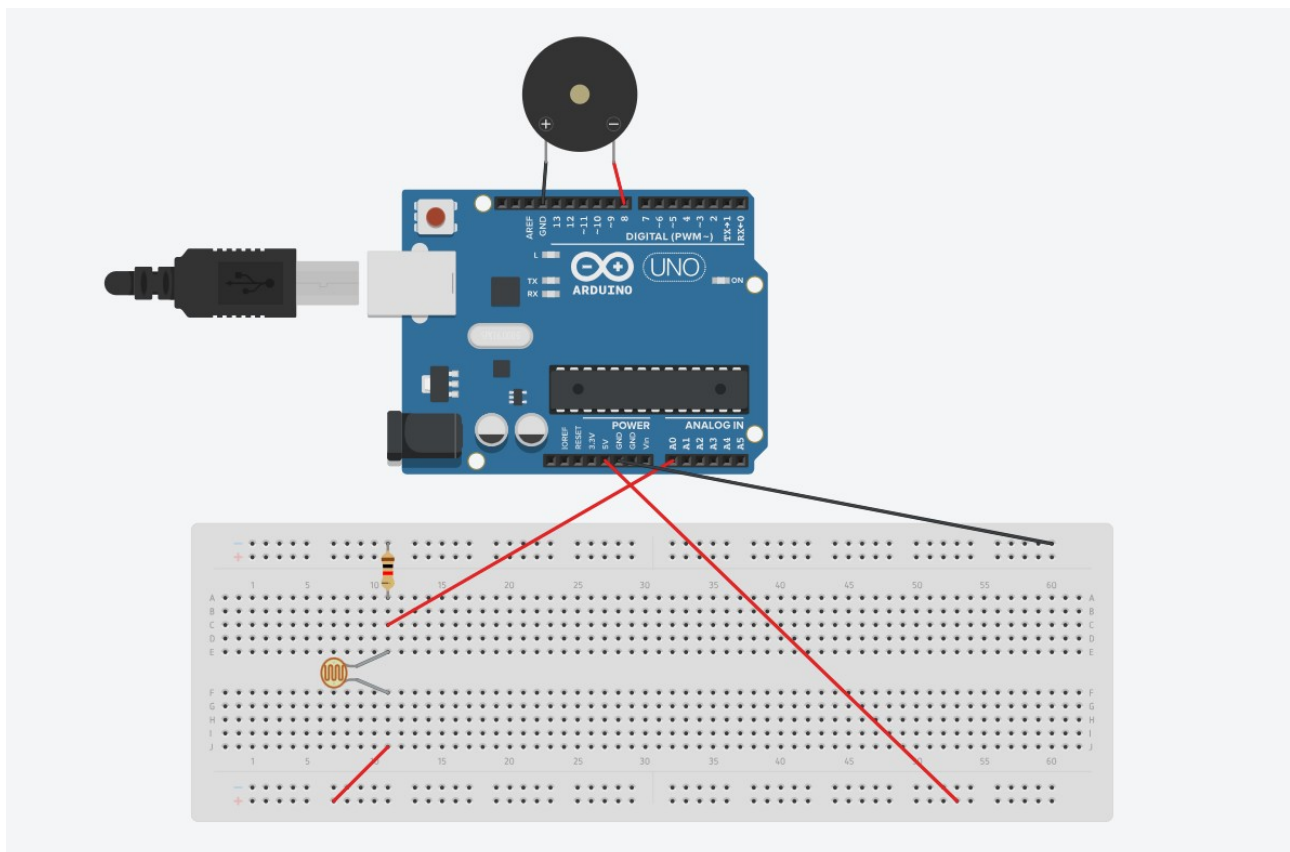
void setup() {
}

void loop() {
  int songLength = sizeof(melody) / sizeof(melody[0]);
  for (int thisNote = 0; thisNote < songLength; thisNote++) {
    // pin8 output the voice, every scale is 0.5 sencond
    tone(8, melody[thisNote], duration);
    // Output the voice after several minutes
    delay(125);
  }

  // restart after two seconds
  delay(2000);
}

```

Theremín sin leds



```

// variable to hold sensor value
int sensorValue;
// variable to calibrate low value

```

```

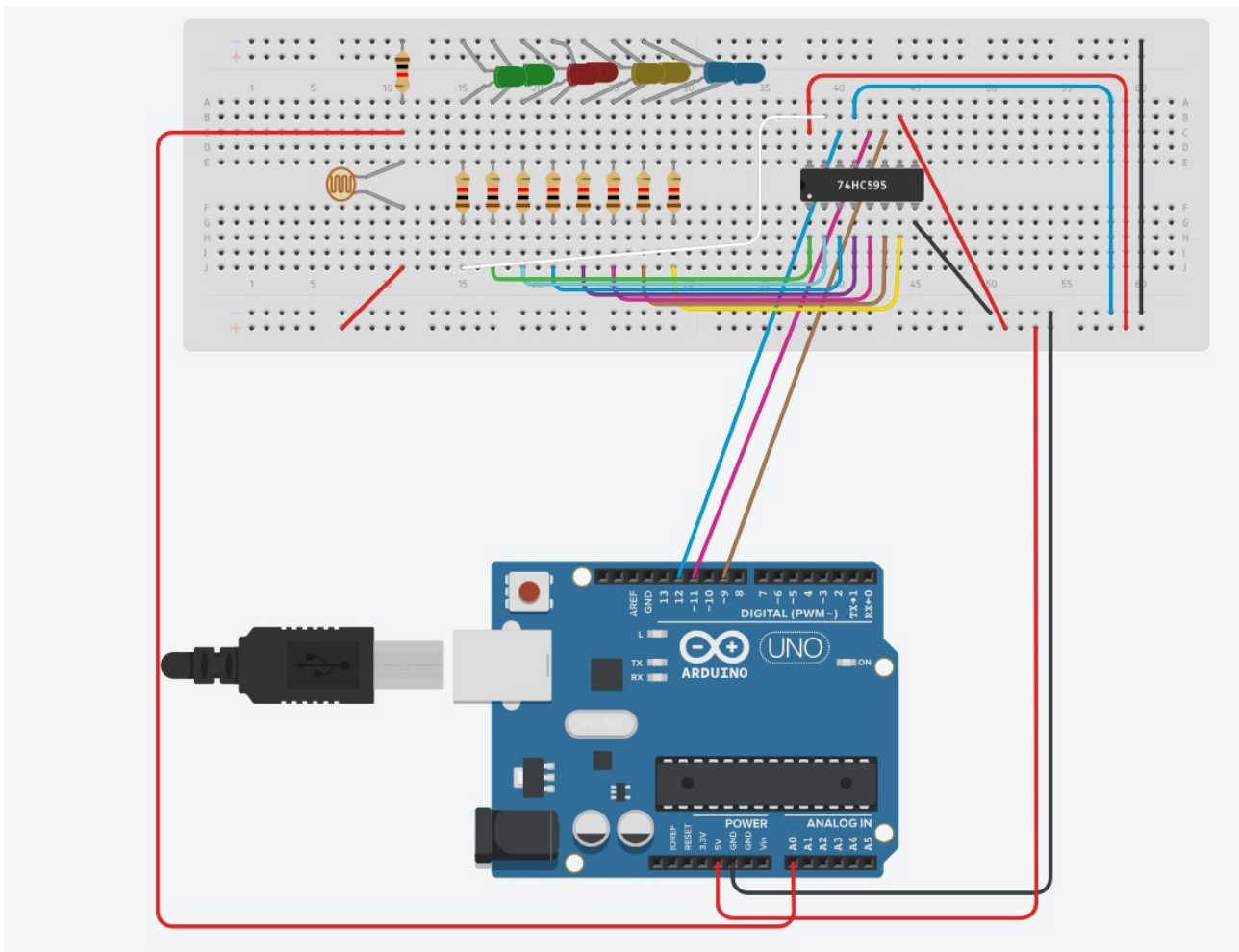
int sensorLow = 1023;
// variable to calibrate high value
int sensorHigh = 0;
// LED pin
const int ledPin = 13;

void setup() {
  // Make the LED pin an output and turn it on
  Serial.begin(9600);
  pinMode(ledPin, OUTPUT);
  digitalWrite(ledPin, HIGH);
  // calibrate for the first five seconds after program runs
  while (millis() < 5000) {
    // record the maximum sensor value
    sensorValue = analogRead(A0);
    if (sensorValue > sensorHigh) {
      sensorHigh = sensorValue;
    }
    // record the minimum sensor value
    if (sensorValue < sensorLow) {
      sensorLow = sensorValue;
    }
  }
  // turn the LED off, signaling the end of the calibration period
  digitalWrite(ledPin, LOW);
}

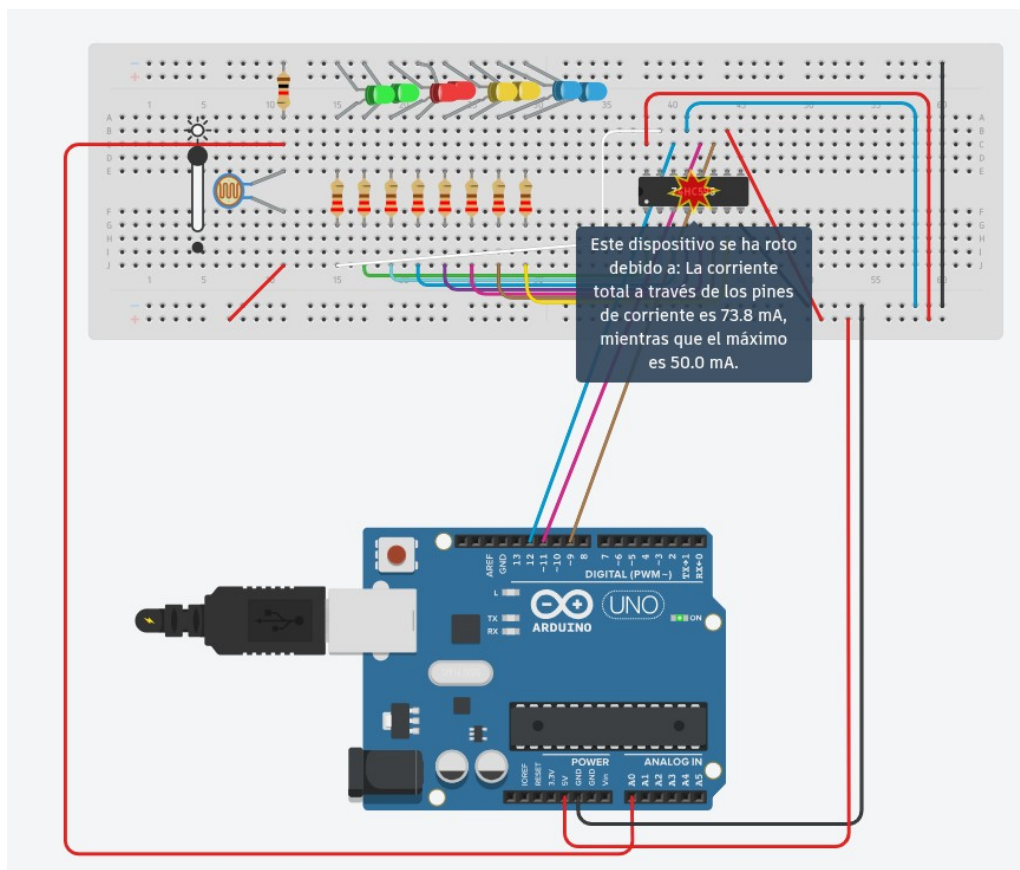
void loop() {
  //read the input from A0 and store it in a variable
  sensorValue = analogRead(A0);
  Serial.print("Value: ");
  Serial.print(sensorValue);
  Serial.print(" | Low: ");
  Serial.print(sensorLow);
  Serial.print(" | High: ");
  Serial.println(sensorHigh);
  // map the sensor values to a wide range of pitches
  int pitch = map(sensorValue, sensorLow, sensorHigh, 50, 4000);
  // play the tone for 20 ms on pin 8
  tone(8, pitch, 20);
  // wait for a moment
  delay(10);
}

```

Theremín con leds



Tuve que cambiar las resistencias de 220ohmios por las de 1kohmio, ya que daba error



```

int lightPin = 0;
int latchPin = 11;
int clockPin = 9;
int dataPin = 12;

int leds = 0;

void setup()
{
  pinMode(latchPin, OUTPUT);
  pinMode(dataPin, OUTPUT);
  pinMode(clockPin, OUTPUT);
}
void updateShiftRegister()
{
  digitalWrite(latchPin, LOW);
  shiftOut(dataPin, clockPin, LSBFIRST, leds);
  digitalWrite(latchPin, HIGH);
}
void loop()
{
  int reading = analogRead(lightPin);
  int numLEDSLit = reading / 57; //1023 / 9 / 2
  if (numLEDSLit > 8) numLEDSLit = 8;
  leds = 0; // no LEDs lit to start
  for (int i = 0; i < numLEDSLit; i++)
  {
    leds = leds + (1 << i);
  }
  updateShiftRegister();
}

```